

Availability

Definition

Availability is the percentage of time a system is operational, accepts requests, and returns the expected response.

It answers:

- Is the system up?
- Can users access it now?
- Is it responding successfully?

Common Availability Targets (SLA)

Availability	Downtime / Year	Downtime / Month	Downtime / Week	Downtime / Day
99.9% (Three 9s)	8.77 hours	43.8 mins	10.1 mins	1.44 mins
99.99% (Four 9s)	52.6 mins	4.38 mins	1.01 mins	8.64 sec
99.999% (Five 9s)	5.26 mins	26.3 sec	6.05 sec	864 ms

Why Availability Matters

High availability is critical for:

- Banking systems
- E-commerce websites
- Social media platforms
- Healthcare systems
- Payment systems
- SaaS products

If unavailable:

- Revenue loss
- User frustration
- Reputation damage
- SLA penalties

Techniques to Achieve High Availability

1. Redundancy

Use multiple servers instead of one.

Example:

- If one server fails, another handles traffic.

2. Load Balancer

Distributes traffic across healthy servers.

Benefits:

- Removes single point of failure
- Better uptime

3. Replication

Maintain copies of data across multiple machines.

Types:

- Primary-Replica
- Multi-master
- Cross-region replication

4. Multi-Region Deployment

Deploy in different geographic regions.

Example:

- India region fails → Singapore region serves traffic.

5. Auto Failover

Automatically redirect traffic when service/node fails.

6. Health Checks

Continuously verify service health.

Examples:

- `/health`
- DB connectivity
- CPU overload detection

7. Monitoring + Alerting

Track outages quickly.

Metrics:

- Errors
- Latency
- CPU
- Memory
- Request failures

Tools:

- Prometheus
- Grafana
- Datadog
- New Relic

Tradeoffs of High Availability

Improving availability may increase:

- Cost
- Complexity
- Operational overhead
- Eventual consistency issues
- Network replication cost

Availability vs Consistency (CAP Theorem)

In distributed systems during partition:

Choose between:

- **Availability** → respond always
- **Consistency** → latest correct data

Examples:

Prefer Availability:

- Social media feed
- Product catalog
- Search engine

Prefer Consistency:

- Bank balance
- Payments
- Ticket booking

Example

Banking Transfer

May sacrifice availability to maintain consistency.

Video Streaming

Prefer availability even if recommendations stale.

Incident Metrics

MTTR (Mean Time To Recovery)

Average time to restore service.

Lower is better.

MTBF (Mean Time Between Failures)

Average time between incidents.

Higher is better.

Interview Notes

When discussing availability ask:

1. Is 24/7 uptime required?
2. Allowed downtime?
3. Single region or multi-region?
4. Failover RTO/RPO?
5. Can stale data be served?

Real-World Examples

Netflix

Uses multi-region active-active architecture.

Amazon Web Services

Provides highly available managed services.

One-Line Memory Trick

Availability = Keep system running even when failures happen.